

Enadraturi Gauss

$$\int_a^b f(x) \cdot w(x) dx \approx ? \quad \int_a^b f(x) \cdot w(x) dx = \underbrace{\sum_{k=1}^n A_k \cdot f(x_k)}_{I_n} + R_n(f)$$

Input:

- $f = \text{fct. integrabil}$
- $a, b = \text{capetele } (-\infty \leq a < b \leq \infty)$
- $w = \text{fct. pondere } (w \geq 0, w \text{ cont. pe } (a, b))$
- $n = \text{nr. de noduri}$

Output:

- I_n
- noduri: $[x_1, \dots, x_n]$
- coeficienți: $[A_1, \dots, A_n]$

a. n. $R_n(p) = 0, \forall p \in \mathcal{P}_{2n-1}$
 sp. pol. de gr. $\leq 2n-1$
 gr. de exactitate

Notatie: $(f(x), g(x))_w := \int_a^b f(x) g(x) w(x) dx$

• $x_1, \dots, x_n$ sunt rădăcinile pol. monic (coef. dominant = 1)
ortogonal π_n în rap. cu w ($\pi_n(x) = (x-x_1) \dots (x-x_n)$)

$$A_k = \int_a^b \frac{(x-x_1) \dots (x-x_{k-1})(x-x_{k+1}) \dots (x-x_n)}{(x-x_1) \dots (x-x_{k-1})(x-x_{k+1}) \dots (x-x_n)} \cdot w(x) dx = \int_a^b \frac{\pi_n(x)}{(x-x_k) \cdot \pi_n'(x_k)} w(x) dx,$$

• pol. monic ortogonal în rap. cu w
 sunt unic determinate de: $k=1, n.$

$$\pi_{k+1}(x) = (x - \lambda_k) \pi_k(x) - \beta_k \pi_{k-1}(x), \quad k=0, 1, 2, \dots$$

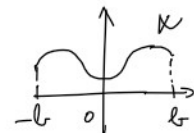
$$\pi_{-1}(x) = 0, \quad \pi_0(x) = 1$$

$$\alpha_k = \frac{(x \pi_k(x), \pi_k(x))_w}{(\pi_k(x), \pi_k(x))_w}, \quad k=0, 1, 2, \dots$$

$$\beta_k = \frac{(\pi_k(x), \pi_{k-1}(x))_w}{(\pi_{k-1}(x), \pi_{k-1}(x))_w}, \quad k=1, 2, \dots, \quad \beta_0 := \int_a^b w(x) dx.$$

Obs: $a = -b$ și $w(-x) = w(x), x \in (-b, b)$

$$\Rightarrow \begin{cases} \alpha_k = 0 \\ \pi_k(-x) = (-1)^k \pi_k(x) \end{cases}, \quad k=0, 1, 2, \dots$$



$$\Rightarrow \begin{cases} x_{n+1-l_2} = -x_{l_2} \\ A_{n+1-l_2} = A_{l_2} \end{cases}, \quad l_2 = \overline{1, n}.$$

Ex $\therefore w(x) = \frac{1}{\sqrt{1-x^2}}, x \in (-1, 1) \rightarrow$ Chebyshev I

$$x_k = \cos\left(\frac{2k-1}{2n} \cdot \pi\right), \quad A_k = \frac{1}{n}, \quad l_2 = \overline{1, n}.$$

Alg. numeric:

$$J_n(w) = \begin{bmatrix} \alpha_0 & \sqrt{\beta_1} & & & 0 \\ \sqrt{\beta_1} & \alpha_1 & \sqrt{\beta_2} & & \\ & \sqrt{\beta_2} & \ddots & \ddots & \\ 0 & & \ddots & \sqrt{\beta_{n-1}} & \alpha_{n-1} \end{bmatrix}$$

$$[V, x] = \text{eig}(J, 'vector')$$

- $x \rightarrow$ nodurile
- $V \rightarrow$ extrag prima linie
 $\rightarrow V(1, :)$
 $\rightarrow A = \beta_0 \cdot V(1, :).12.$

$w(t)$	Support	Name	α_k	β_0	$\beta_k, k \geq 1$
1	$[-1, 1]$	Legendre	0	2	$1/(4-k^2)$
1	$[0, 1]$	Shifted Legendre	$\frac{1}{2}$	1	$1/(4(4-k^2))$
$(1-t^2)^{-1/2}$	$[-1, 1]$	Chebyshev #1	0	π	$\frac{1}{2} (k=1), \frac{1}{4} (k>1)$
$(1-t^2)^{1/2}$	$[-1, 1]$	Chebyshev #2	0	$\frac{1}{2}\pi$	$\frac{1}{4}$
$(1-t)^{-1/2}(1+t)^{1/2}$	$[-1, 1]$	Chebyshev #3	$\frac{1}{2} (k=0)$ $0 (k>0)$	π	$\frac{1}{4}$
$(1-t)^{1/2}(1+t)^{-1/2}$	$[-1, 1]$	Chebyshev #4	$-\frac{1}{2} (k=0)$ $0 (k>0)$	π	$\frac{1}{4}$
$(1-t^2)^{\lambda-1/2}, \lambda > -\frac{1}{2}$	$[-1, 1]$	Gegenbauer	0	$\sqrt{\pi} \frac{\Gamma(\lambda+\frac{1}{2})}{\Gamma(\lambda+1)}$	$\frac{k(k+2\lambda-1)}{4(k+\lambda)(k+\lambda-1)}$
$(1-t)^\alpha(1+t)^\beta$ $\alpha > -1, \beta > -1$	$[-1, 1]$	Jacobi	α_k^J	β_0^J	β_k^J
e^{-t}	$[0, \infty]$	Laguerre	$2k+1$	1	k^2
$t^\alpha e^{-t}, \alpha > -1$	$[0, \infty]$	Generalized Laguerre	$2k+\alpha+1$	$\Gamma(1+\alpha)$	$k(k+\alpha)$
e^{-t^2}	$[-\infty, \infty]$	Hermite	0	$\sqrt{\pi}$	$\frac{1}{2}k$
$ t ^{2\mu} e^{-t^2}, \mu > -\frac{1}{2}$	$[-\infty, \infty]$	Generalized Hermite	0	$\Gamma(\mu + \frac{1}{2})$	$\frac{1}{2}k (k \text{ even})$ $\frac{1}{2}k + \mu (k \text{ odd})$
$\frac{1}{2\pi} e^{(2\phi-\pi)t} \Gamma(\lambda+it) ^2$ $\lambda > 0, 0 < \phi < \pi$	$[-\infty, \infty]$	Meixner-Pollaczek	$-\frac{k+\lambda}{\tan \phi}$	$\frac{\Gamma(2\lambda)}{(2 \sin \phi)^{2\lambda}}$	$\frac{k(k+2\lambda-1)}{4 \sin^2 \phi}$

$$\alpha_k^J = \frac{\beta^2 - \alpha^2}{(2k+\alpha+\beta)(2k+\alpha+\beta+2)}^*$$

$$\beta_0^J = \frac{2^{\alpha+\beta+1} \Gamma(\alpha+1) \Gamma(\beta+1)}{\Gamma(\alpha+\beta+2)}, \quad \beta_k^J = \frac{4k(k+\alpha)(k+\beta)(k+\alpha+\beta)}{(2k+\alpha+\beta)^2(2k+\alpha+\beta+1)(2k+\alpha+\beta-1)}^\dagger$$

*If $k=0$, the common factor $\alpha+\beta$ in the numerator and denominator of α_0^J should be (must be, if $\alpha+\beta=0$) cancelled.

†If $k=1$, the last factors in the numerator and denominator of β_1^J should be (must be, if $\alpha+\beta+1=0$) cancelled.

The first few Legendre polynomials are:

n	$P_n(x)$
0	1
1	x
2	$\frac{1}{2} (3x^2 - 1)$
3	$\frac{1}{2} (5x^3 - 3x)$
4	$\frac{1}{8} (35x^4 - 30x^2 + 3)$
5	$\frac{1}{8} (63x^5 - 70x^3 + 15x)$
6	$\frac{1}{16} (231x^6 - 315x^4 + 105x^2 - 5)$
7	$\frac{1}{16} (429x^7 - 693x^5 + 315x^3 - 35x)$
8	$\frac{1}{128} (6435x^8 - 12012x^6 + 6930x^4 - 1260x^2 + 35)$
9	$\frac{1}{128} (12155x^9 - 25740x^7 + 18018x^5 - 4620x^3 + 315x)$
10	$\frac{1}{256} (46189x^{10} - 109395x^8 + 90090x^6 - 30030x^4 + 3465x^2 - 63)$

monic
 1
 x
 $x^2 - \frac{1}{3}$
 $x^3 - \frac{3}{5}x$
 $\vdots$

The first few Chebyshev polynomials of the first kind are [OEIS: A028297](#)

$$\begin{aligned}
 T_0(x) &= 1 \\
 T_1(x) &= x \\
 T_2(x) &= 2x^2 - 1 \\
 T_3(x) &= 4x^3 - 3x \\
 T_4(x) &= 8x^4 - 8x^2 + 1 \\
 T_5(x) &= 16x^5 - 20x^3 + 5x \\
 T_6(x) &= 32x^6 - 48x^4 + 18x^2 - 1 \\
 T_7(x) &= 64x^7 - 112x^5 + 56x^3 - 7x \\
 T_8(x) &= 128x^8 - 256x^6 + 160x^4 - 32x^2 + 1 \\
 T_9(x) &= 256x^9 - 576x^7 + 432x^5 - 120x^3 + 9x \\
 T_{10}(x) &= 512x^{10} - 1280x^8 + 1120x^6 - 400x^4 + 50x^2 - 1 \\
 T_{11}(x) &= 1024x^{11} - 2816x^9 + 2816x^7 - 1232x^5 + 220x^3 - 11x
 \end{aligned}$$

The first few Chebyshev polynomials of the second kind are [OEIS: A053117](#)

$$U_0(x) = 1$$

$$U_1(x) = 2x$$

$$U_2(x) = 4x^2 - 1$$

$$U_3(x) = 8x^3 - 4x$$

$$U_4(x) = 16x^4 - 12x^2 + 1$$

$$U_5(x) = 32x^5 - 32x^3 + 6x$$

$$U_6(x) = 64x^6 - 80x^4 + 24x^2 - 1$$

$$U_7(x) = 128x^7 - 192x^5 + 80x^3 - 8x$$

$$U_8(x) = 256x^8 - 448x^6 + 240x^4 - 40x^2 + 1$$

$$U_9(x) = 512x^9 - 1024x^7 + 672x^5 - 160x^3 + 10x$$

These are the first few Laguerre polynomials: $\alpha = 0, \quad \mathcal{L}(x) = e^{-x}$

n	$L_n(x)$
0	1
1	$-x + 1$
2	$\frac{1}{2}(x^2 - 4x + 2)$
3	$\frac{1}{6}(-x^3 + 9x^2 - 18x + 6)$
4	$\frac{1}{24}(x^4 - 16x^3 + 72x^2 - 96x + 24)$
5	$\frac{1}{120}(-x^5 + 25x^4 - 200x^3 + 600x^2 - 600x + 120)$
6	$\frac{1}{720}(x^6 - 36x^5 + 450x^4 - 2400x^3 + 5400x^2 - 4320x + 720)$
n	$\frac{1}{n!}((-x)^n + n^2(-x)^{n-1} + \dots + n(n!)(-x) + n!)$

The first eleven physicist's Hermite polynomials are:

$$H_0(x) = 1,$$

$$H_1(x) = 2x,$$

$$H_2(x) = 4x^2 - 2,$$

$$H_3(x) = 8x^3 - 12x,$$

$$H_4(x) = 16x^4 - 48x^2 + 12,$$

$$H_5(x) = 32x^5 - 160x^3 + 120x,$$

$$H_6(x) = 64x^6 - 480x^4 + 720x^2 - 120,$$

$$H_7(x) = 128x^7 - 1344x^5 + 3360x^3 - 1680x,$$

$$H_8(x) = 256x^8 - 3584x^6 + 13440x^4 - 13440x^2 + 1680,$$

$$H_9(x) = 512x^9 - 9216x^7 + 48384x^5 - 80640x^3 + 30240x,$$

$$H_{10}(x) = 1024x^{10} - 23040x^8 + 161280x^6 - 403200x^4 + 302400x^2 - 30240.$$

Formule pentru resturi în cuadraturi gaussiene

• **Gauss:**

$$\int_a^b f(x)w(x)dx - \sum_{i=1}^n A_i \cdot f(x_i) = \frac{f^{(2n)}(\xi)}{(2n)!} \int_a^b \pi_n^2(x) \cdot w(x)dx$$

→ gr. de exact. $2n-1$

π_n = polinomul ortogonal (în raport cu ponderea w) de grad n care este monic (coeficientul lui x^n este 1, i.e. $\pi_n(x) = (x - x_1) \cdot \dots \cdot (x - x_n)$).

• **Gauss-Radau:**

$$\int_a^b f(x)w(x)dx - A_1 \cdot f(a) - \sum_{i=2}^n A_i \cdot f(x_i) = \frac{f^{(2n-1)}(\xi)}{(2n-1)!} \int_a^b \pi_{n-1}^2(x) \cdot (x-a) \cdot w(x)dx$$

$$\int_a^b f(x)w(x)dx - \sum_{i=1}^{n-1} A_i \cdot f(x_i) - A_n \cdot f(b) = \frac{f^{(2n-1)}(\xi)}{(2n-1)!} \int_a^b \pi_{n-1}^2(x) \cdot (b-x) \cdot w(x)dx$$

→ gr. exact: $2n-2$

π_{n-1} = polinomul ortogonal (în raport cu ponderea $w_a(x) = (x-a)w(x)$,

respectiv $w_b(x) = (b-x)w(x)$) de grad $n-1$ care este monic (coeficientul lui x^{n-1} este 1, i.e. $\pi_{n-1}(x) = (x - x_2) \cdot \dots \cdot (x - x_n)$, respectiv $\pi_{n-1}(x) = (x - x_1) \cdot \dots \cdot (x - x_{n-1})$).

• **Gauss-Lobatto:**

$$\int_a^b f(x)w(x)dx - A_1 \cdot f(a) - \sum_{i=2}^{n-1} A_i \cdot f(x_i) - A_n \cdot f(b) = \frac{f^{(2n-2)}(\xi)}{(2n-2)!} \int_a^b \pi_{n-2}^2(x) \cdot (x-a) \cdot (b-x) \cdot w(x)dx$$

→ gr. ex.: $2n-3$

π_{n-2} = polinomul ortogonal (în raport cu ponderea $w_{a,b}(x) = (x-a)(b-x)w(x)$) de grad $n-2$ care este monic (coeficientul lui x^{n-2} este 1, i.e. $\pi_{n-2}(x) = (x - x_2) \cdot \dots \cdot (x - x_{n-1})$).

Problemă:

a) Găsiți o formulă de cuadratură de forma

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx = A_1 f(-1) + A_2 f(x_2) + A_3 f(x_3) + A_4 f(1) + R(f),$$

determinând coeficienții și nodurile necunoscute astfel încât să aibă gradul maxim de exactitate. Determinați formula restului.

b) Aplicați formula de cuadratură de mai sus pentru a estima integrala

$$\int_{-1}^1 \frac{e^x}{\sqrt{1-x^2}} dx.$$

Găsiți o margine superioară a erorii de aproximare, folosind formula restului.

Rez.: a) Aceasta este o cuadratură Gauss-Lobatto (are primul și ultimul nod prestabilite - capetele intervalului).

```

syms x; a=-sym(1); b=sym(1);
w=1/sqrt(1-x^2);
wab=simplify((x-a)*(b-x)*w);
pi2=orto_poly_sym_type('Cebisev2',2);
sols=solve(pi2,x);

nodes=[a sols' b]
coefs=gauss_coefs_sym(w,a,b,nodes)

rest_fara_f=1/factorial(6)*int(pi2^2*wab,x,a,b)
w = (sym)

```

$$\frac{1}{\sqrt{1-x^2}}$$

```

wab = (sym)

```

$$\sqrt{1-x^2}$$

```

nodes = (sym) [-1 -1/2 1/2 1] (1x4 matrix)
coefs = (sym 1x4 matrix)
rest_fara_f = (sym)

```

```

pi
-----
23040

```

```

b) IG=eval(coefs*exp(nodes)')
eroare_max=eval(rest_fara_f)*exp(1)

IO1=integral(@(x) exp(x)./sqrt(1-x.^2),-1,1)
IO2=quad(@(x) exp(x)./sqrt(1-x.^2),-1,1)

```

```

IG = 3.977604560937058
eroare_max = 3.706481867479846e-04
IO1 = 3.977462976085627
IO2 = 3.977463260506118

```